

OrbitRS Protocol Comparison Whitepaper

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OrbitRS Protocol Comparison Whitepaper

Comprehensive Feature-by-Feature and Command-by-Command Analysis

Version: 1.0 **Date:** December 2024 **Authors:** OrbitRS Development Team

Executive Summary

OrbitRS is a revolutionary multi-protocol database engine written in Rust that natively implements **8 database protocols** from a single unified storage layer. This whitepaper provides an exhaustive feature-by-feature and command-by-command comparison across all supported protocols, enabling architects and developers to make informed decisions about which protocol interfaces best suit their workloads.

Key Statistics: - **8 Native Protocols:** PostgreSQL, MySQL, Redis (RESP), CQL (Cassandra), Cypher (Neo4j), MongoDB, OrbitQL, REST/gRPC - **350+ Commands:** Comprehensive command coverage across all protocols - **5 Data Models:** Relational, Document, Key-Value, Graph, Time-Series - **1 Unified Storage:** All protocols share the same underlying data

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1. Protocol Overview

1.1 Supported Protocols

Protocol	Port	Wire Protocol	Primary Use Case	Status
PostgreSQL	5432	PostgreSQL v3.0	ACID transactions, complex queries	Production
MySQL	3306	MySQL 4.1+	Web applications, LAMP stack	Production
Redis (RESP)	6379	RESP2/RESP3	Caching, real-time data, pub/sub	Production
CQL (Cassandra)	9042	CQL v4	Time-series, high-write throughput	Production
Cypher (Neo4j)	7687	Bolt v4/v5	Graph traversal, relationships	Production
MongoDB	27017	MongoDB Wire	Flexible documents, rapid prototyping	Production
OrbitQL	5432	PostgreSQL Wire	Multi-model queries, AI/ML integration	Production
REST API	8080	HTTP/WebSocket	Universal access, microservices	Production
gRPC	50051	Protocol Buffers	High-performance RPC, actor management	Production

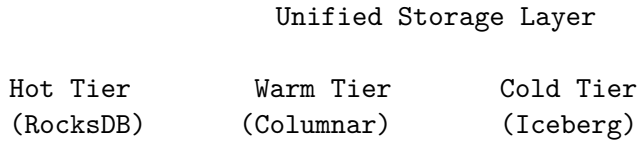
1.2 Protocol Hierarchy

OrbitRS Protocol Stack

PostgreSQL :5432	MySQL :3306	Redis :6379	CQL :9042	Cypher :7687
MongoDB :27017	OrbitQL :5432	REST :8080		

Unified Query Engine

SQL Engine	Graph Engine	KV Engine	TS Engine
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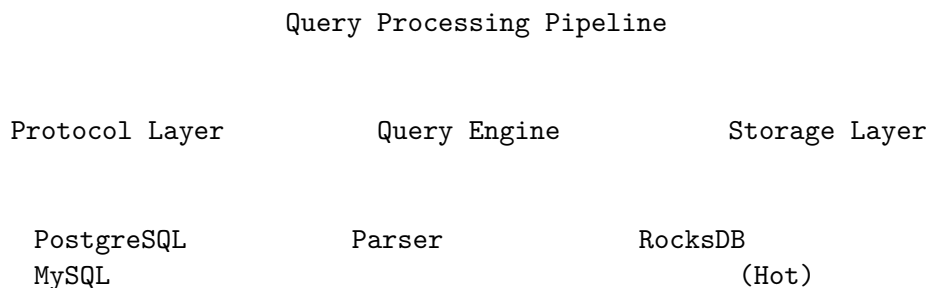


2. Architecture Comparison

2.1 Protocol Implementation Depth

Protocol	Implementation Level	Native Features	OrbitRS Extensions
PostgreSQL	Full wire protocol v3.0	MVCC, prepared statements, COPY	pgvector, JSON/JSONB
MySQL	Full wire protocol 4.1+	Prepared statements, multi-result	Batch optimization
Redis	Complete RESP2/RESP3	All data structures, pub/sub	Vector search, GraphRAG
CQL	Full CQL v4 protocol	Batch, TTL, consistency levels	ANN search, vector types
Cypher	Bolt v4/v5 protocol	Pattern matching, procedures	GraphRAG integration
MongoDB	OP_MSG protocol	BSO, cursors, aggregation	Vector search
OrbitQL	Native (PostgreSQL wire)	Graph + SQL + Vector	ML inference, LIVE queries
REST	OpenAPI compliant	CRUD, WebSocket, SSE	Actor management, CDC

2.2 Query Processing Pipeline



Redis	Analyzer	
CQL		Columnar
Cypher	Optimizer	(Warm)
MongoDB		
OrbitQL	Executor	Iceberg
REST		(Cold)

3. Feature Comparison Matrix

3.1 Core Database Features

Feature	PostgreSQL	MySQL	Redis	CQL	Cypher	MongoDB	OrbitQL
ACID	Full	Full			Full	Per-doc	Full
Transac-tions			MULTI	Tun-able			
MVCC							
Joins	All types	All types			Pattern	\$lookup	All + Graph
Aggregations	Full	Full		Lim-ited		Pipeline	Full
Subqueries			Limited				
CTEs							
(WITH)							
Window						\$setWin-dowFields	
Func-tions							
Prepared							
State-ments							
Stored			Lua			\$function	
Proce-dures				UDF			
Triggers						Change Streams	

3.2 Data Type Support

Data Type	PostgreSQL	MySQL	Redis	CQL	Cypher	MongoDB	OrbitQL
Integer							
Float/Decimal							
String/Text							

Data Type	PostgreSQL	MySQL	Redis	CQL	Cypher	MongoDB	OrbitQL
Boolean							
Date/Time							
UUID							
JSON/JSONB					Map	BSON	
Arrays			List				
Maps/Objects	JSONB		Hash				
Sets							
Vectors	pgvector		VEC-TOR.*				
Geospatial	PostGIS		GEO*				
Binary/Blob							

3.3 Advanced Features

Feature	PostgreSQL	MySQL	Redis	CQL	Cypher	MongoDB	OrbitQL
Graph					Native		Native
Traversal			GRAPH.*			\$graphLookup	
Vector	pgvector		VEC-TOR.*	ANN		\$vectorSearch	Native
Search			FT.*	SAI		\$text	
Full-Text							
Search							
Time-Series			TS.*	Na-tive		Time-Series	
Pub/Sub	LIS-TEN/NOTIFY		Native			Change Streams	LIVE
Streaming			XREAD				LIVE
ML Inte-gration							ml:.*
GraphRAG			GRAPHRAG.*				Native

4. Command Reference by Protocol

4.1 Redis (RESP) Commands (150+ Commands)

String Commands (31 commands)

Command	Description	Example
GET	Get string value	GET key
SET	Set string value	SET key value [EX seconds]
MGET	Get multiple values	MGET key1 key2 key3
MSET	Set multiple values	MSET k1 v1 k2 v2

Command	Description	Example
INCR	Increment integer	INCR counter
DECR	Decrement integer	DECR counter
INCRBY	Increment by amount	INCRBY counter 5
APPEND	Append to string	APPEND key " suffix"
STRLEN	Get string length	STRLEN key
GETRANGE	Get substring	GETRANGE key 0 10
SETEX	Set with expiry	SETEX key 3600 value
SETNX	Set if not exists	SETNX key value
GETSET	Get old, set new	GETSET key newvalue
GETDEL	Get and delete	GETDEL key
GETEX	Get and set expiry	GETEX key EX 100

Hash Commands (11 commands)

Command	Description	Example
HGET	Get field value	HGET hash field
HSET	Set field value	HSET hash field value
HMGET	Get multiple fields	HMGET hash f1 f2
HMSET	Set multiple fields	HMSET hash f1 v1 f2 v2
HGETALL	Get all fields	HGETALL hash
HDEL	Delete field	HDEL hash field
HEXISTS	Check field exists	HEXISTS hash field
HKEYS	Get all field names	HKEYS hash
HVALS	Get all values	HVALS hash
HLEN	Get field count	HLEN hash
HINCRBY	Increment field	HINCRBY hash field 1

List Commands (14 commands)

Command	Description	Example
LPUSH	Push to head	LPUSH list value
RPUSH	Push to tail	RPUSH list value
LPOP	Pop from head	LPOP list
RPOP	Pop from tail	RPOP list
LRANGE	Get range	LRANGE list 0 -1
LLEN	Get length	LLEN list
LINDEX	Get by index	LINDEX list 0
LSET	Set by index	LSET list 0 value
LRM	Remove elements	LRM list 1 value
LTRIM	Trim list	LTRIM list 0 99
LINSERT	Insert element	LINSERT list BEFORE pivot value
BLPOP	Blocking pop head	BLPOP list 0
BRPOP	Blocking pop tail	BRPOP list 0

Set Commands (8 commands)

Command	Description	Example
SADD	Add member	SADD set member
SREM	Remove member	SREM set member
SMEMBERS	Get all members	SMEMBERS set
SCARD	Get cardinality	SCARD set
SISMEMBER	Check membership	SISMEMBER set member
SUNION	Union sets	SUNION set1 set2
SINTER	Intersect sets	SINTER set1 set2
SDIFF	Difference sets	SDIFF set1 set2

Sorted Set Commands (10 commands)

Command	Description	Example
ZADD	Add with score	ZADD zset 1.0 member
ZREM	Remove member	ZREM zset member
ZSCORE	Get score	ZSCORE zset member
ZRANK	Get rank	ZRANK zset member
ZRANGE	Get range by rank	ZRANGE zset 0 10
ZRANGEBYSCORE	Get range by score	ZRANGEBYSCORE zset 0 100
ZCARD	Get cardinality	ZCARD zset
ZCOUNT	Count in range	ZCOUNT zset 0 100
ZINCRBY	Increment score	ZINCRBY zset 1.0 member

Stream Commands (14 commands)

Command	Description	Example
XADD	Add to stream	XADD stream * field value
XREAD	Read from stream	XREAD STREAMS stream 0
XREADGROUP	Read as consumer group	XREADGROUP GROUP g consumer STREAMS s >
XRANGE	Get range	XRANGE stream - +
XREVRANGE	Get reverse range	XREVRANGE stream + -
XLEN	Get length	XLEN stream
XINFO	Get stream info	XINFO STREAM stream
XGROUP	Manage groups	XGROUP CREATE stream group \$
XACK	Acknowledge message	XACK stream group id
XCLAIM	Claim message	XCLAIM stream group consumer 0 id
XPENDING	Get pending	XPENDING stream group

Command	Description	Example
XTRIM	Trim stream	XTRIM stream MAXLEN 1000
XDEL	Delete entry	XDEL stream id

Vector Commands (12 commands)

Command	Description	Example
VECTOR.CREATE	Create index	VECTOR.CREATE idx DIM 128 METRIC COSINE
VECTOR.ADD	Add vector	VECTOR.ADD idx id [0.1, 0.2, ...]
VECTOR.GET	Get vector	VECTOR.GET idx id
VECTOR.DEL	Delete vector	VECTOR.DEL idx id
VECTOR.SEARCH	Search vectors	VECTOR.SEARCH idx [0.1, ...] K 10
VECTOR.INFO	Get index info	VECTOR.INFO idx
VECTOR.COUNT	Count vectors	VECTOR.COUNT idx
VECTOR.LIST	List indices	VECTOR.LIST
VECTOR.STATS	Get statistics	VECTOR.STATS idx
VECTOR.DROP	Drop index	VECTOR.DROP idx
FT.CREATE	Create FT index	FT.CREATE idx ON HASH ...
FT.SEARCH	Full-text search	FT.SEARCH idx "query"

Time Series Commands (14 commands)

Command	Description	Example
TS.CREATE	Create series	TS.CREATE ts:temp RETENTION 86400
TS.ADD	Add data point	TS.ADD ts:temp * 25.5
TS.GET	Get latest	TS.GET ts:temp
TS.RANGE	Get range	TS.RANGE ts:temp - + AGGREGATION avg 60
TS.MRANGE	Multi-series range	TS.MRANGE - + FILTER sensor=temp
TS.ALTER	Alter series	TS.ALTER ts:temp RETENTION 172800
TS.INFO	Get info	TS.INFO ts:temp
TS.MADD	Add multiple	TS.MADD ts:temp * 25 ts:hum * 60
TS.INCRBY	Increment	TS.INCRBY ts:counter 1
TS.DECRBY	Decrement	TS.DECRBY ts:counter 1

Command	Description	Example
TS.CREATERULE	Create rule	TS.CREATERULE src dst AGGREGATION avg 60
TS.DELETERULE	Delete rule	TS.DELETERULE src dst
TS.QUERYINDEX	Query index	TS.QUERYINDEX sensor=*

Graph Commands (8 commands)

Command	Description	Example
GRAPH.QUERY	Execute query	GRAPH.QUERY g "MATCH (n) RETURN n"
GRAPH.RO_QUERY	Read-only query	GRAPH.RO_QUERY g "MATCH (n) RETURN n"
GRAPH.DELETE	Delete graph	GRAPH.DELETE g
GRAPH.EXPLAIN	Explain query	GRAPH.EXPLAIN g "MATCH (n) RETURN n"
GRAPH.PROFILE	Profile query	GRAPH.PROFILE g "MATCH (n) RETURN n"
GRAPH.CONFIG	Get/set config	GRAPH.CONFIG GET MAX_QUEUED_QUERIES
GRAPH.SLOWLOG	Get slow queries	GRAPH.SLOWLOG g

GraphRAG Commands (7 commands)

Command	Description	Example
GRAPHRAG.INDEX	Index documents	GRAPHRAG.INDEX idx doc1 "content"
GRAPHRAG.SEARCH	Search with RAG	GRAPHRAG.SEARCH idx "query" K 5
GRAPHRAG.CHAT	Chat with context	GRAPHRAG.CHAT idx "question"
GRAPHRAG.RETRIEVE	Retrieve context	GRAPHRAG.RETRIEVE idx "query"
GRAPHRAG.DELETE	Delete document	GRAPHRAG.DELETE idx doc1
GRAPHRAG.LIST	List indices	GRAPHRAG.LIST
GRAPHRAG.INFO	Get index info	GRAPHRAG.INFO idx

Pub/Sub Commands (6 commands)

Command	Description	Example
PUBLISH	Publish message	PUBLISH channel message
SUBSCRIBE	Subscribe	SUBSCRIBE channel

Command	Description	Example
UNSUBSCRIBE	Unsubscribe	UNSUBSCRIBE channel
PSUBSCRIBE	Pattern subscribe	PSUBSCRIBE pattern*
PUNSUBSCRIBE	Pattern unsub	PUNSUBSCRIBE pattern*
PUBSUB	Pub/sub info	PUBSUB CHANNELS

Transaction Commands (5 commands)

Command	Description	Example
MULTI	Start transaction	MULTI
EXEC	Execute transaction	EXEC
DISCARD	Discard transaction	DISCARD
WATCH	Watch keys	WATCH key1 key2
UNWATCH	Unwatch keys	UNWATCH

4.2 PostgreSQL SQL Statements (80+ Statement Types)

Data Definition Language (DDL)

Statement	Description	Example
CREATE DATABASE	Create database	CREATE DATABASE mydb
CREATE TABLE	Create table	CREATE TABLE users (id SERIAL PRIMARY KEY, name TEXT)
CREATE INDEX	Create index	CREATE INDEX idx_name ON users(name)
CREATE VIEW	Create view	CREATE VIEW active_users AS SELECT * FROM users WHERE active
CREATE SCHEMA	Create schema	CREATE SCHEMA myschema
CREATE FUNCTION	Create function	CREATE FUNCTION add(a INT, b INT) RETURNS INT AS \$\$... \$\$
CREATE TRIGGER	Create trigger	CREATE TRIGGER tr AFTER INSERT ON users ...
CREATE SEQUENCE	Create sequence	CREATE SEQUENCE user_id_seq
CREATE TYPE	Create type	CREATE TYPE status AS ENUM ('active', 'inactive')

Statement	Description	Example
CREATE EXTENSION	Create extension	CREATE EXTENSION vector
ALTER TABLE	Modify table	ALTER TABLE users ADD COLUMN email TEXT
ALTER SEQUENCE	Modify sequence	ALTER SEQUENCE user_id_seq RESTART WITH 100
DROP TABLE	Drop table	DROP TABLE users CASCADE
DROP INDEX	Drop index	DROP INDEX idx_name
DROP VIEW	Drop view	DROP VIEW active_users
TRUNCATE	Truncate table	TRUNCATE users RESTART IDENTITY

Data Manipulation Language (DML)

Statement	Description	Example
SELECT	Query data	SELECT * FROM users WHERE age > 18
INSERT	Insert data	INSERT INTO users (name) VALUES ('John')
UPDATE	Update data	UPDATE users SET name = 'Jane' WHERE id = 1
DELETE	Delete data	DELETE FROM users WHERE id = 1
UPSERT	Insert/update	INSERT INTO users ... ON CONFLICT DO UPDATE
COPY	Bulk load	COPY users FROM '/data/users.csv' CSV

Query Clauses

Clause	Description	Example
WHERE	Filter rows	WHERE age > 18 AND active = true
GROUP BY	Group rows	GROUP BY department
HAVING	Filter groups	HAVING COUNT(*) > 5
ORDER BY	Sort results	ORDER BY created_at DESC
LIMIT/OFFSET	Paginate	LIMIT 10 OFFSET 20

Clause	Description	Example
DISTINCT	Unique rows	SELECT DISTINCT category FROM products
JOIN	Join tables	LEFT JOIN orders ON users.id = orders.user_id
UNION	Combine results	SELECT ... UNION SELECT ...
WITH (CTE)	Common table expr	WITH cte AS (SELECT ...) SELECT * FROM cte

Transaction Control

Statement	Description	Example
BEGIN	Start transaction	BEGIN TRANSACTION
COMMIT	Commit transaction	COMMIT
ROLLBACK	Rollback transaction	ROLLBACK
SAVEPOINT	Create savepoint	SAVEPOINT sp1
RELEASE	Release savepoint	RELEASE SAVEPOINT sp1

Window Functions

Function	Description	Example
ROW_NUMBER()	Row number	ROW_NUMBER() OVER (ORDER BY id)
RANK()	Rank with gaps	RANK() OVER (ORDER BY score DESC)
DENSE_RANK()	Rank without gaps	DENSE_RANK() OVER (ORDER BY score DESC)
LAG()	Previous row	LAG(value) OVER (ORDER BY date)
LEAD()	Next row	LEAD(value) OVER (ORDER BY date)
SUM() OVER	Running sum	SUM(amount) OVER (ORDER BY date)
AVG() OVER	Running average	AVG(amount) OVER (ROWS 7 PRECEDING)

pgvector Operations

Operation	Description	Example
<=>	Cosine distance	embedding <=> '[0.1,0.2,0.3]'

Operation	Description	Example
<->	L2 distance	<code>embedding <-></code> <code>'[0.1,0.2,0.3]'</code>
<#>	Inner product	<code>embedding <#></code> <code>'[0.1,0.2,0.3]'</code>
<code>vector()</code>	Create vector	<code>vector('[0.1,0.2,0.3]')</code>
<code>CREATE INDEX ... USING ivfflat</code>	IVF index	<code>CREATE INDEX ON items</code> <code>USING ivfflat</code> <code>(embedding)</code>
<code>CREATE INDEX ... USING hns</code>	HNSW index	<code>CREATE INDEX ON items</code> <code>USING hns</code> <code>(embedding)</code>

4.3 MySQL Wire Protocol Commands

Command Code	Name	Description
0x00	COM_SLEEP	Sleep
0x01	COM_QUIT	Close connection
0x02	COM_INIT_DB	Select database
0x03	COM_QUERY	Execute SQL
0x04	COM_FIELD_LIST	List fields
0x05	COM_CREATE_DB	Create database
0x06	COM_DROP_DB	Drop database
0x07	COM_REFRESH	Refresh
0x09	COM_STATISTICS	Get statistics
0x0E	COM_PING	Ping server
0x16	COM_STMT_PREPARE	Prepare statement
0x17	COM_STMT_EXECUTE	Execute prepared
0x18	COM_STMT_SEND_LONG_DATA	Send blob data
0x19	COM_STMT_CLOSE	Close prepared
0x1A	COM_STMT_RESET	Reset prepared
0x1B	COM_SET_OPTION	Set option
0x1C	COM_STMT_FETCH	Fetch rows
0x1F	COM_RESET_CONNECTION	Reset connection

4.4 CQL (Cassandra) Statements (45+ Operations)

Schema Operations

Statement	Description	Example
CREATE KEYSPACE	Create keyspace	CREATE KEYSPACE ks WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 3}
CREATE TABLE	Create table	CREATE TABLE users (id UUID PRIMARY KEY, name TEXT)
CREATE INDEX	Create index	CREATE INDEX ON users (name)
CREATE TYPE	Create UDT	CREATE TYPE address (street TEXT, city TEXT)
CREATE MATERIALIZED VIEW	Create MV	CREATE MATERIALIZED VIEW users_by_email AS SELECT * FROM users WHERE email IS NOT NULL PRIMARY KEY (email, id)
CREATE FUNCTION	Create UDF	CREATE FUNCTION my_func(input TEXT) RETURNS TEXT LANGUAGE java AS '...'
CREATE AGGREGATE	Create aggregate	CREATE AGGREGATE my_agg(INT) SFUNC state_func STYPE INT
ALTER TABLE	Modify table	ALTER TABLE users ADD email TEXT
ALTER KEYSPACE	Modify keyspace	ALTER KEYSPACE ks WITH replication = {...}
DROP TABLE	Drop table	DROP TABLE users
DROP KEYSPACE	Drop keyspace	DROP KEYSPACE ks
TRUNCATE	Truncate table	TRUNCATE users
USE	Select keyspace	USE mykeyspace

Data Operations

Statement	Description	Example
SELECT	Query data	SELECT * FROM users WHERE id = ?
INSERT	Insert data	INSERT INTO users (id, name) VALUES (uuid(), 'John')

Statement	Description	Example
UPDATE	Update data	UPDATE users SET name = 'Jane' WHERE id = ?
DELETE	Delete data	DELETE FROM users WHERE id = ?
BATCH	Batch operations	BEGIN BATCH INSERT ... UPDATE ... APPLY BATCH

CQL-Specific Features

Feature	Description	Example
TTL	Time-to-live	INSERT INTO users ... USING TTL 86400
TIMESTAMP	Write timestamp	INSERT INTO users ... USING TIMESTAMP 1234567890
IF NOT EXISTS	Conditional insert	INSERT INTO users ... IF NOT EXISTS
IF EXISTS	Conditional update	UPDATE users SET ... IF EXISTS
ALLOW FILTERING	Allow full scan	SELECT * FROM users WHERE age > 18 ALLOW FILTERING
PER PARTITION LIMIT	Limit per partition	SELECT * FROM events PER PARTITION LIMIT 10
JSON	JSON format	SELECT JSON * FROM users
DISTINCT	Distinct partition keys	SELECT DISTINCT user_id FROM events
Counter operations	Counter columns	UPDATE counters SET count = count + 1 WHERE id = ?
Collection operations	List/Set/Map ops	UPDATE users SET tags = tags + {'new_tag'} WHERE id = ?

Consistency Levels

Level	Description
ONE	One replica
TWO	Two replicas
THREE	Three replicas
QUORUM	Majority of replicas

Level	Description
ALL	All replicas
LOCAL_QUORUM	Quorum in local DC
EACH_QUORUM	Quorum in each DC
LOCAL_ONE	One in local DC
ANY	Any node (including hints)
SERIAL	Linearizable (for LWT)
LOCAL_SERIAL	Local linearizable

4.5 Cypher (Neo4j) Query Language

Reading Clauses

Clause	Description	Example
MATCH	Pattern match	MATCH (n:Person)-[:KNOWS]->(m)
OPTIONAL MATCH	Left outer match	OPTIONAL MATCH (n)-[:OWNS]->(p)
WHERE	Filter	WHERE n.age > 18
RETURN	Return results	RETURN n.name, count(m)
WITH	Intermediate results	WITH n, count(m) AS friends
UNWIND	Expand list	UNWIND [1,2,3] AS x
ORDER BY	Sort	ORDER BY n.name DESC
SKIP	Skip rows	SKIP 10
LIMIT	Limit rows	LIMIT 25

Writing Clauses

Clause	Description	Example
CREATE	Create nodes/rels	CREATE (n:Person {name: 'John'})
MERGE	Create if not exists	MERGE (n:Person {id: 1})
SET	Set properties	SET n.name = 'Jane'
DELETE	Delete nodes/rels	DELETE n
DETACH DELETE	Delete with rels	DETACH DELETE n
REMOVE	Remove property	REMOVE n.temp_field
FOREACH	Iterate and update	FOREACH (x IN list CREATE (:Item {val: x}))

Pattern Syntax

Pattern	Description	Example
(n)	Any node	<code>MATCH (n)</code>
(n:Label)	Labeled node	<code>MATCH (n:Person)</code>
(n {prop: val})	Node with property	<code>MATCH (n {name: 'John'})</code>
-->	Outgoing relationship	<code>(a)-->(b)</code>
<--	Incoming relationship	<code>(a)<--(b)</code>
--	Any direction	<code>(a)--(b)</code>
-[:TYPE]->	Typed relationship	<code>(a)-[:KNOWS]->(b)</code>
-[r:TYPE]->	Named relationship	<code>(a)-[r:KNOWS]->(b)</code>
-[*1..3]->	Variable length	<code>(a)-[:KNOWS*1..3]->(b)</code>
-[:TYPE1 TYPE2]->	Multiple types	<code>(a)-[:KNOWS LIKES]->(b)</code>

Aggregation Functions

Function	Description	Example
<code>count()</code>	Count	<code>count(n)</code>
<code>sum()</code>	Sum	<code>sum(n.amount)</code>
<code>avg()</code>	Average	<code>avg(n.score)</code>
<code>min()</code>	Minimum	<code>min(n.age)</code>
<code>max()</code>	Maximum	<code>max(n.age)</code>
<code>collect()</code>	Collect to list	<code>collect(n.name)</code>
<code>percentileCont()</code>	Percentile	<code>percentileCont(n.score, 0.5)</code>
<code>stDev()</code>	Standard deviation	<code>stDev(n.value)</code>

Graph Algorithms (via Procedures)

Procedure	Description	Example
<code>gds.shortestPath.dijkstra</code>	Shortest path	<code>CALL</code> <code>gds.shortestPath.dijkstra.stream(...)</code>
<code>gds.pageRank</code>	PageRank	<code>CALL</code> <code>gds.pageRank.stream(...)</code>
<code>gds.louvain</code>	Community detection	<code>CALL</code> <code>gds.louvain.stream(...)</code>
<code>gds.betweenness</code>	Betweenness centrality	<code>CALL</code> <code>gds.betweenness.stream(...)</code>
<code>gds.nodeSimilarity</code>	Node similarity	<code>CALL</code> <code>gds.nodeSimilarity.stream(...)</code>

4.6 MongoDB Operations

CRUD Operations

Operation	Description	Example
<code>find</code>	Query documents	<code>db.users.find({age: {\$gt: 18}})</code>
<code>findOne</code>	Query single doc	<code>db.users.findOne({_id: ObjectId(...)})</code>
<code>insertOne</code>	Insert document	<code>db.users.insertOne({name: 'John'})</code>
<code>insertMany</code>	Insert multiple	<code>db.users.insertMany([{...}, {...}])</code>
<code>updateOne</code>	Update document	<code>db.users.updateOne({_id: ...}, {\$set: {...}})</code>
<code>updateMany</code>	Update multiple	<code>db.users.updateMany({active: false}, {\$set: {...}})</code>
<code>replaceOne</code>	Replace document	<code>db.users.replaceOne({_id: ...}, {...})</code>
<code>deleteOne</code>	Delete document	<code>db.users.deleteOne({_id: ...})</code>
<code>deleteMany</code>	Delete multiple	<code>db.users.deleteMany({active: false})</code>

Query Operators

Operator	Description	Example
<code>\$eq</code>	Equal	<code>{age: {\$eq: 25}}</code>
<code>\$ne</code>	Not equal	<code>{status: {\$ne: 'deleted'}}</code>
<code>\$gt</code>	Greater than	<code>{age: {\$gt: 18}}</code>
<code>\$gte</code>	Greater or equal	<code>{age: {\$gte: 18}}</code>
<code>\$lt</code>	Less than	<code>{age: {\$lt: 65}}</code>
<code>\$lte</code>	Less or equal	<code>{age: {\$lte: 65}}</code>
<code>\$in</code>	In array	<code>{status: {\$in: ['active', 'pending']}}</code>
<code>\$nin</code>	Not in array	<code>{status: {\$nin: ['deleted']}}</code>
<code>\$and</code>	Logical AND	<code>{\$and: [{age: {\$gte: 18}}, {active: true}]}</code>
<code>\$or</code>	Logical OR	<code>{\$or: [{age: {\$lt: 18}}, {status: 'minor'}]}</code>
<code>\$not</code>	Logical NOT	<code>{age: {\$not: {\$lt: 18}}}</code>
<code>\$exists</code>	Field exists	<code>{email: {\$exists: true}}</code>
<code>\$regex</code>	Regular expression	<code>{name: {\$regex: /^John/}}</code>

Aggregation Pipeline

Stage	Description	Example
<code>\$match</code>	Filter	<code>{\$match: {status: 'active'}}</code>

Stage	Description	Example
\$group	Group	<code>{ \$group: { _id: '\$category', count: { \$sum: 1 } } }</code>
\$project	Projection	<code>{ \$project: { name: 1, _id: 0 } }</code>
\$sort	Sort	<code>{ \$sort: { createdAt: -1 } }</code>
\$limit	Limit	<code>{ \$limit: 10 }</code>
\$skip	Skip	<code>{ \$skip: 20 }</code>
\$unwind	Flatten array	<code>{ \$unwind: '\$tags' }</code>
\$lookup	Join	<code>{ \$lookup: { from: 'orders', ... } }</code>
\$graphLookup	Recursive lookup	<code>{ \$graphLookup: { from: 'employees', ... } }</code>
\$vectorSearch	Vector search	<code>{ \$vectorSearch: { vector: [...], path: 'embedding', ... } }</code>

4.7 OrbitQL Statements (Native Multi-Model)

SQL-Compatible Statements

Statement	Description	Example
SELECT	Query data	<code>SELECT * FROM users WHERE age > 18</code>
INSERT	Insert data	<code>INSERT INTO users { name: 'John', age: 25 }</code>
UPDATE	Update data	<code>UPDATE users SET age = 26 WHERE name = 'John'</code>
DELETE	Delete data	<code>DELETE FROM users WHERE id = 1</code>
CREATE	Create schema	<code>CREATE TABLE users (id INT, name STRING)</code>
DROP	Drop schema	<code>DROP TABLE users</code>
ALTER	Alter schema	<code>ALTER TABLE users ADD COLUMN email STRING</code>

Graph Operations (SurrealDB-style)

Statement	Description	Example
RELATE	Create relationship	RELATE users:john->follows->users:jane
TRAVERSE	Graph traversal	SELECT ->follows->user FROM users:john
MATCH	Pattern matching	MATCH (n:User)-[:FOLLOWS]->(m)

Real-time Queries

Statement	Description	Example
LIVE SELECT	Subscribe to changes	LIVE SELECT * FROM orders WHERE status = 'pending' DIFF
KILL	Stop subscription	KILL \$subscription_id

Schema Definition (DEFINE)

Statement	Description	Example
DEFINE NAMESPACE	Define namespace	DEFINE NAMESPACE myapp
DEFINE DATABASE	Define database	DEFINE DATABASE production
DEFINE TABLE	Define table	DEFINE TABLE users SCHEMALESS
DEFINE FIELD	Define field	DEFINE FIELD email ON users TYPE string
DEFINE INDEX	Define index	DEFINE INDEX email_idx ON users FIELDS email UNIQUE

ML Integration

Function	Description	Example
ml::predict()	Run prediction	SELECT ml::predict('churn_model', {...}) FROM users
ml::embed_text()	Text embedding	SELECT ml::embed_text('hello world')

Function	Description	Example
<code>ml::classify()</code>	Classification	SELECT ml::classify('sentiment', text) FROM reviews
<code>ml::cluster()</code>	Clustering	SELECT ml::cluster('kmeans', features) FROM data

Graph Functions

Function	Description	Example
<code>graph::shortest_path()</code>	Shortest path	SELECT graph::shortest_path(a, b) FROM nodes
<code>graph::neighbors()</code>	Get neighbors	SELECT graph::neighbors(node, 2) FROM nodes
<code>graph::connected()</code>	Check connected	SELECT graph::connected(a, b) FROM nodes
<code>graph::pagerank()</code>	PageRank	SELECT graph::pagerank() FROM nodes

Vector Operations

Syntax	Description	Example
<code><=></code>	Cosine similarity	<code>embedding <=></code> <code>\$query_vector</code>
<code>vector::similarity::cosine()</code>	Cosine function	<code>vector::similarity::cosine(a, b)</code>
<code>vector::similarity::euclidean()</code>	Euclidean distance	<code>vector::similarity::euclidean(a, b)</code>
<code>vector::similarity::dot()</code>	Dot product	<code>vector::similarity::dot(a, b)</code>

4.8 REST API Endpoints

Actor Management

Method	Endpoint	Description
GET	<code>/api/v1/actors</code>	List actors (paginated)

Method	Endpoint	Description
GET	/api/v1/actors/{id}	Get actor state
POST	/api/v1/actors	Create actor
PUT	/api/v1/actors/{id}	Update actor
DELETE	/api/v1/actors/{id}	Deactivate actor
POST	/api/v1/actors/{id}/invoke	Invoke method

Query Endpoints

Method	Endpoint	Description
POST	/api/v1/query	Execute query
GET	/api/v1/query/history	Query history
POST	/api/v1/query/explain	Explain query

Transaction Management

Method	Endpoint	Description
POST	/api/v1/transactions	Begin transaction
POST	/api/v1/transactions/{id}/commit	Commit
POST	/api/v1/transactions/{id}/abort	Abort

Cluster Information

Method	Endpoint	Description
GET	/api/v1/cluster/status	Cluster status
GET	/api/v1/cluster/nodes	List nodes
GET	/api/v1/cluster/config	Configuration

Real-time

Method	Endpoint	Description
WS	/api/v1/ws/actors/{id}	Actor events
WS	/api/v1/ws/events	System events
GET	/api/v1/stream/query	SSE query stream
GET	/api/v1/stream/cdc	CDC stream

5. Data Model Comparison

5.1 Schema Flexibility

Protocol	Schema Model	Flexibility	Best For
PostgreSQL	Schema-on-write	Rigid	Structured data, ACID
MySQL	Schema-on-write	Rigid	Web applications
Redis	Schemaless	Maximum	Caching, sessions
CQL	Schema-on-write	Semi-rigid	Time-series, IoT
Cypher	Schema-optional	Flexible	Relationships
MongoDB	Schema-on-read	Flexible	Documents, prototyping
OrbitQL	Hybrid	Configurable	Multi-model

5.2 Relationship Modeling

Approach	PostgreSQL	Redis	CQL	Cypher	MongoDB	OrbitQL
Foreign Keys	Native					
Joins	All types			Pattern	\$lookup	All + Graph
Embedded Docs	JSONB	Hash		Props	Native	
Graph Edges				Native		RELATE
Denormalization	Manual	GRAPH.* Native	Native		Native	Hybrid

5.3 Query Patterns

Query Pattern Suitability		
Point Lookup (key=value)	PostgreSQL Redis CQL MongoDB	(optimal)
Range Scan (key > x)	PostgreSQL Redis CQL MongoDB	
Complex Joins (multi-table)	PostgreSQL Redis CQL OrbitQL	(optimal)
Graph Traversal (relationships)	PostgreSQL Cypher OrbitQL	(optimal)

	Redis	(GRAPH.*)
Full-Text Search	PostgreSQL Redis MongoDB OrbitQL	(FT.*)
Vector Similarity	PostgreSQL Redis CQL OrbitQL	(pgvector) (VECTOR.*) (ANN) (native)
Time-Series	Redis CQL OrbitQL	(TS.*) (optimal)

6. Query Capabilities Comparison

6.1 Query Language Features

Feature	PostgreSQL	MySQL	Redis	CQL	Cypher	MongoDB	OrbitQL
Declarative SQL				SQL-like			
Imperative Commands	PL/pgSQL			UDF			
Pattern Matching	LIKE	LIKE			Native		
Recursive Queries	CTE	CTE			*1..n		
Parameterized Prepared Statements						\$graphLookup	
Explain Plans							

6.2 Aggregation Capabilities

Function	PostgreSQL	MySQL	Redis	CQL	Cypher	MongoDB	OrbitQL
COUNT							
SUM							

Function	PostgreSQL	MySQL	Redis	CQL	Cypher	MongoDB	OrbitQL
AVG							
MIN/MAX							
GROUP_CONCAT					collect	\$push	
STDDEV							
PERCENTILE							
DISTINCT							
ROLLUP							
CUBE							

7. Transaction Support Comparison

7.1 ACID Properties

Property	PostgreSQL	MySQL	Redis	CQL	Cypher	MongoDB	OrbitQL
Atomicity	Full	Full			Full	Doc	Full
Consistency	Full	Full	MULTI	Batch	Full		Full
				Even- tual			
Isolation	4 levels	4 levels				Snapshot	4 levels
Durability	Full	Full					Full
			AOF/RDB				

7.2 Isolation Levels

Level	PostgreSQL	MySQL	OrbitQL	Description
READ UNCOMMITTED				Dirty reads allowed
READ COMMITTED	Default		Default	No dirty reads
REPEATABLE READ		Default		No phantom reads
SERIALIZABLE				Full isolation

7.3 Distributed Transaction Support

Feature	PostgreSQL	Redis	CQL	Cypher	OrbitQL
2-Phase Commit					
Saga Pattern					
Distributed Locks	Advisory	SETNX	LWT		
Deadlock Detection					

8. Performance Characteristics

8.1 Latency Comparison

Operation Type	PostgreSQL	MySQL	Redis	CQL	Cypher	MongoDB	OrbitQL
Point Read	1-5ms	1-5ms	<1ms	1-5ms	5-20ms	1-5ms	1-5ms
Point Write	5-20ms	5-20ms	<1ms	1-5ms	10-50ms	2-10ms	5-20ms
Range Scan	5-50ms	5-50ms	5-20ms	5-30ms	10-100ms	5-50ms	5-50ms
Complex Join	10-500ms	10-500ms	N/A	N/A	20-200ms	20-200ms	10-500ms
Aggregation	10-1000ms	10-1000ms	10-50ms	10-500ms	20-500ms	10-500ms	10-500ms
Graph Traversal	N/A	N/A	5-50ms	N/A	5-100ms	20-200ms	5-100ms
Vector Search	5-50ms	N/A	5-20ms	5-50ms	N/A	10-50ms	5-20ms

8.2 Throughput Characteristics

Protocol	Read TPS	Write TPS	Best Workload
PostgreSQL	50K+	10K+	OLTP, complex queries
MySQL	50K+	10K+	Web applications
Redis	1M+	500K+	Caching, sessions
CQL	100K+	500K+	Time-series, logs
Cypher	10K+	5K+	Graph analytics
MongoDB	100K+	50K+	Documents
OrbitQL	50K+	10K+	Multi-model

8.3 Scalability

Scalability Characteristics

Horizontal Scale (add nodes)	PostgreSQL	(with extensions)
	Redis	(Cluster)
	CQL	(native)
	MongoDB	(sharding)
	OrbitQL	(Raft cluster)
Vertical Scale (bigger node)	PostgreSQL	
	MySQL	
	Redis	

	All	
Multi-Region	CQL	(native)
	MongoDB	
	OrbitQL	(planned)

9. Use Case Recommendations

9.1 Protocol Selection Guide

Use Case	Recommended Protocol	Reason
User Sessions	Redis	Sub-ms latency, TTL support
Financial Transactions	PostgreSQL/OrbitQL	Full ACID, audit trail
Product Catalog	MongoDB/PostgreSQL	Flexible schema, rich queries
Social Network	Cypher/OrbitQL	Native graph traversal
IoT Time-Series	CQL/Redis TS	High write throughput
Real-time Analytics	Redis/OrbitQL	In-memory, pub/sub
Content Management	MongoDB	Document flexibility
E-commerce Orders	PostgreSQL	ACID, complex joins
Recommendation Engine	Cypher + OrbitQL	Graph + ML
Search	Redis FT/PostgreSQL FTS	Full-text indexing
Caching	Redis	Speed, data structures
AI/ML Applications	OrbitQL	Native ML integration

9.2 Multi-Protocol Patterns

Pattern 1: CQRS with Protocol Separation

Write Path: PostgreSQL (ACID transactions)

Read Path: Redis (cached queries) + Cypher (graph queries)

Pattern 2: Event Sourcing

Event Store: CQL (append-only, partitioned by time)

Projections: PostgreSQL (materialized views)

Cache: Redis (hot data)

Pattern 3: Microservices Integration

Service A	PostgreSQL (orders)	
Service B	MongoDB (products)	Unified Storage
Service C	Redis (sessions)	
Service D	Cypher (recommendations)	

10. Migration Guide

10.1 From Single Protocol to Multi-Protocol

Step	Action	Tools
1	Identify data patterns	Query analysis
2	Map to appropriate protocols	This whitepaper
3	Update connection strings	Configuration
4	Migrate schema	DDL scripts
5	Test with existing drivers	Integration tests
6	Gradual traffic migration	Feature flags

10.2 Protocol Compatibility

From	To (OrbitRS)	Compatibility Level
PostgreSQL	PostgreSQL :5432	100% wire compatible
MySQL	MySQL :3306	100% wire compatible
Redis	Redis :6379	100% RESP compatible
Cassandra	CQL :9042	100% CQL v4 compatible
Neo4j	Cypher :7687	Bolt v4/v5 compatible
MongoDB	MongoDB :27017	Wire protocol compatible

10.3 Connection String Examples

PostgreSQL

postgresql://user:pass@orbit-host:5432/mydb

MySQL

```
mysql://user:pass@orbit-host:3306/mydb
```

```
# Redis
```

```
redis://orbit-host:6379
```

```
# Cassandra
```

```
cassandra://orbit-host:9042/keyspace
```

```
# Neo4j (Bolt)
```

```
bolt://orbit-host:7687
```

```
# MongoDB
```

```
mongodb://orbit-host:27017/mydb
```

Appendix: Complete Command Reference

A.1 Command Count by Protocol

Protocol	Category	Command Count
Redis	String	31
	Hash	11
	List	14
	Set	8
	Sorted Set	10
	Stream	14
	Vector	12
	Time Series	14
	Graph	8
	GraphRAG	7
	Pub/Sub	6
	Transaction	5
	Server/Connection	10+
	Total	150+
PostgreSQL	DDL	30+
	DML	6
	TCL	5
	DCL	2
	Total	80+
MySQL	Wire Protocol	18
CQL	Schema	20+
	Data	6
	Total	45+
Cypher	Reading	9
	Writing	7
	Schema	4
	Total	20+

Protocol	Category	Command Count
MongoDB	CRUD	9
	Aggregation	10+
	Total	20+
OrbitQL	SQL	15+
	Graph	5
	ML	4
	Real-time	2
	Total	30+
REST	Endpoints	20+
GRAND TOTAL		350+

Conclusion

OrbitRS provides a unique approach to database architecture by unifying 8 major database protocols under a single storage layer. This whitepaper has demonstrated:

1. **Protocol Depth:** Each protocol is fully implemented with native wire compatibility
2. **Feature Parity:** Advanced features like vector search, graph queries, and ML integration work across protocols
3. **Unified Storage:** All data accessible through any protocol interface
4. **Performance:** Competitive with specialized databases for each workload type

When to Use OrbitRS

- **Polyglot Persistence:** Replace 3-5 databases with one
- **Multi-Model Applications:** Combine SQL, graph, and document queries
- **AI/ML Integration:** Built-in vector search and ML inference
- **Operational Simplicity:** Single backup, single cluster, unified monitoring

Getting Started

```
# Start OrbitRS (all protocols enabled)
./orbit-server --config orbit-server.toml

# Connect with any client
psql -h localhost -p 5432 -U orbit
redis-cli -p 6379
mongosh --port 27017
cqlsh localhost 9042
```

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